

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
import ipywidgets as widgets
from IPython.display import display, clear_output

# Initialize brightness level
brightness = 1.0

def adjust_brightness(change):
    global brightness
    brightness += change # Increase or decrease brightness factor
    brightness = max(0.1, min(brightness, 3.0)) # Keep brightness within range

    led_color = np.array([255, 165, 0]) * brightness # Adjust RGB value for LED
    led_color = np.clip(led_color, 0, 255).astype(np.uint8) # Ensure valid RGB range

    clear_output(wait=True) # Clear previous output
    display_controls()

    plt.figure(figsize=(2, 2))
    plt.imshow([[led_color/255]]) # Display LED with current brightness
    plt.axis("off")
    plt.show()

def display_controls():
    up_button = widgets.Button(description="Increase Brightness")
    down_button = widgets.Button(description="Decrease Brightness")

    up_button.on_click(lambda _: adjust_brightness(0.1))
    down_button.on_click(lambda _: adjust_brightness(-0.1))

    display(up_button, down_button)

# Set up the plot and turn on interactive mode
display_controls()
adjust_brightness(0)
```



Increase Brightness

Decrease Brightness



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